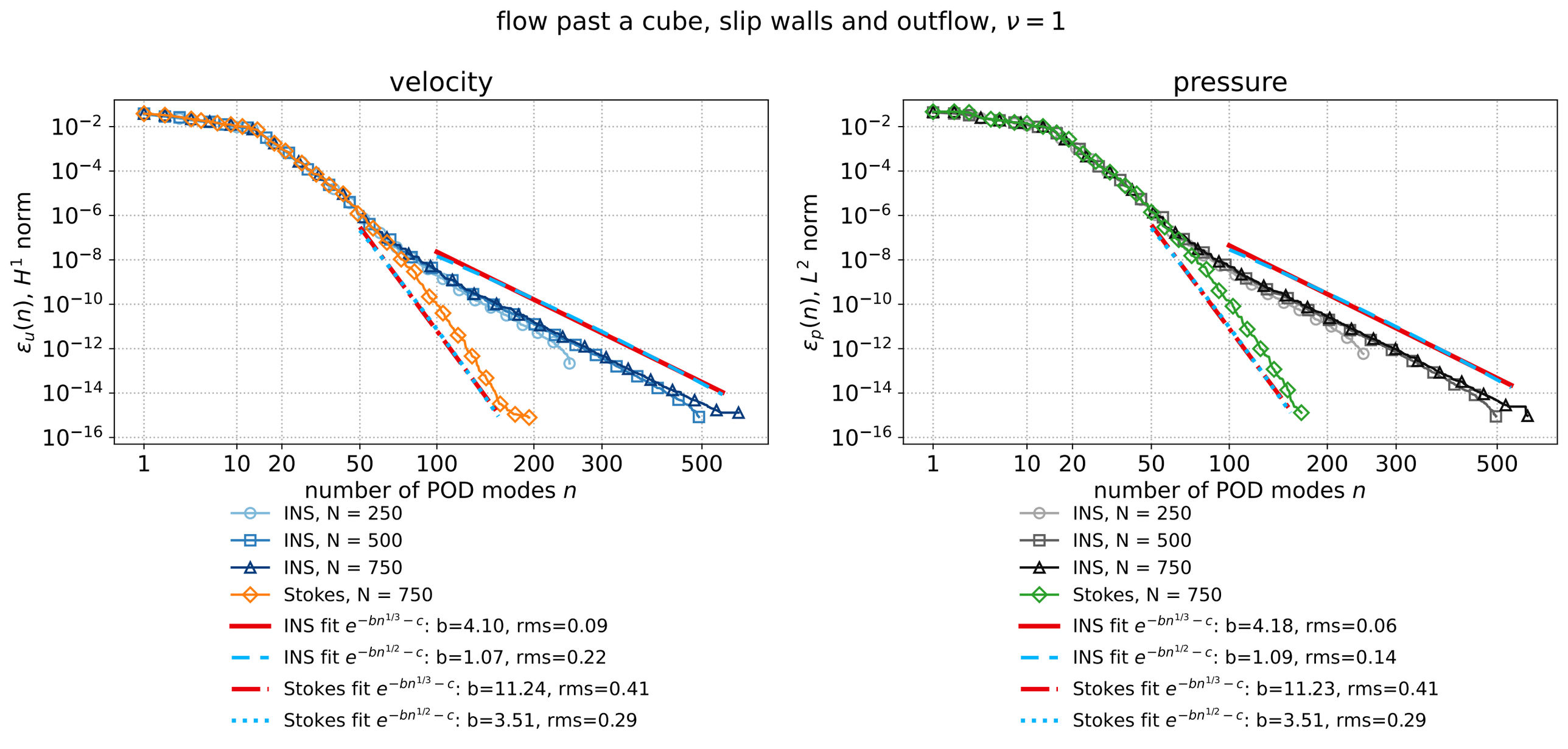
flow past a cube, slip walls and outflow, $\nu = 1$
velocity
pressure
$\varepsilon_u(n)$, H^1 norm
$\varepsilon_p(n)$, L^2 norm
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
1
10
20
50
100
200
300
500
number of POD modes n
INS, N = 250
INS, N = 500
INS, N = 750
Stokes, N = 750
INS fit $e^{-bn^{1/3}-c}$: b=4.10, rms=0.09
INS fit $e^{-bn^{1/2}-c}$: b=1.07, rms=0.22
Stokes fit $e^{-bn^{1/3}-c}$: b=11.24, rms=0.41
Stokes fit $e^{-bn^{1/2}-c}$: b=3.51, rms=0.29
INS fit $e^{-bn^{1/3}-c}$: b=4.18, rms=0.06
INS fit $e^{-bn^{1/2}-c}$: b=1.09, rms=0.14
Stokes fit $e^{-bn^{1/3}-c}$: b=11.23, rms=0.41
Stokes fit $e^{-bn^{1/2}-c}$: b=3.51, rms=0.29